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JEE MAIN · PHYSICS · CHAPTER-WISE PYQS

Waves

JEE Main Physics · Previous Year Questions with Answer Key

176	140	2002 – 2026	41	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

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Numerical 38 questions

Q1. JEE Main 2026 (Online) 5th April Morning Shift

A transverse wave on a string is described by $y = 3\sin(36t + 0.018x + \pi/4)$, where x, y are in cm and t in seconds. The least distance between the two successive crests in the wave is ____ cm . (Nearest integer)

$(\pi = 3.14)$

Open solution

Q2. JEE Main 2026 (Online) 28th January Evening Shift

Two tuning forks *A* and *B* are sounded together giving rise to 8 beats in 2 s. When fork *A* is loaded with wax, the beat frequency is reduced to 4 beats in 2 s. If the original frequency of tuning fork *B* is 380 Hz then original frequency of tuning fork *A* is _____ Hz.

Open solution

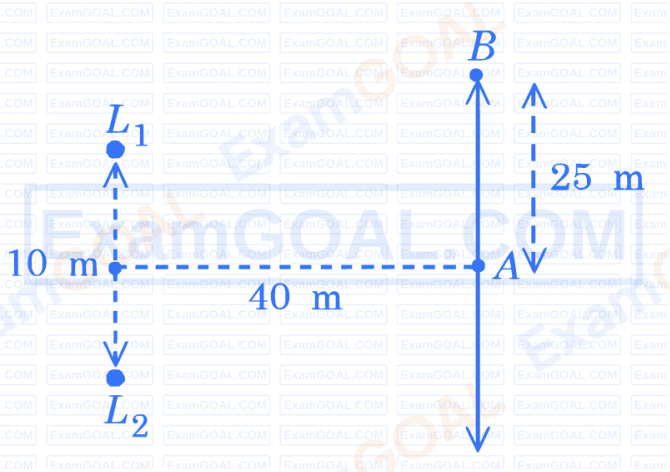
Q3. JEE Main 2026 (Online) 23rd January Evening Shift

The velocity of sound in air is doubled when the temperature is raised from 0°C to $\alpha^\circ\text{C}$. The value of α is ____ .

Open solution

Q4. JEE Main 2026 (Online) 22nd January Morning Shift

Two loudspeakers (L_1 and L_2) are placed with a separation of 10 m , as shown in figure. Both speakers are fed with an audio input signal of same frequency with constant volume. A voice recorder, initially at point A, at equidistant to both loudspeakers, is moved by 25 m along the line AB while monitoring the audio signal. The measured signal was found to undergo 10 cycles of minima and maxima during the movement. The frequency of the input signal is ____ Hz (Speed of sound in air is $324\ \text{m/s}$ and $\sqrt{5} = 2.23$)



Open solution

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Q5. JEE Main 2024 (Online) 8th April Morning Shift

A closed and an open organ pipe have same lengths. If the ratio of frequencies of their seventh overtones is $\left(\frac{a-1}{a}\right)$ then the value of a is _____.

Open solution

Q6. JEE Main 2024 (Online) 6th April Evening Shift

Two open organ pipes of lengths $60\ \text{cm}$ and $90\ \text{cm}$ resonate at 6th and 5th harmonics respectively. The difference of frequencies for the given modes is _____ Hz. (Velocity of sound in air $= 333\ \text{m/s}$)

Open solution

Q7. JEE Main 2024 (Online) 5th April Evening Shift

A sonometer wire of resonating length $90\ \text{cm}$ has a fundamental frequency of $400\ \text{Hz}$ when kept under some tension. The resonating length of the wire with fundamental frequency of $600\ \text{Hz}$ under same tension _____ cm.

Open solution

Q8. JEE Main 2024 (Online) 1st February Morning Shift

A tuning fork resonates with a sonometer wire of length $1\ \text{m}$ stretched with a tension of $6\ \text{N}$. When the tension in the wire is changed to $54\ \text{N}$, the same tuning fork produces 12 beats per second with it. The frequency of the tuning fork is _____ Hz.

Open solution

Q9. JEE Main 2024 (Online) 30th January Evening Shift

A point source is emitting sound waves of intensity $16 \times 10^{-8}\ \text{Wm}^{-2}$ at the origin. The difference in intensity (magnitude only) at two points located at distances of $2m$ and $4m$ from the origin respectively will be _____ $\times 10^{-8}\ \text{Wm}^{-2}$.

Open solution

Q10. JEE Main 2024 (Online) 30th January Morning Shift

In a closed organ pipe, the frequency of fundamental note is $30\ \text{Hz}$. A certain amount of water is now poured in the organ pipe so that the fundamental frequency is increased to $110\ \text{Hz}$. If the organ pipe has a cross-sectional area of $2\ \text{cm}^2$, the amount of water poured in the organ tube is _____ g. (Take speed of sound in air is $330\ \text{m/s}$)

Open solution

Q11. JEE Main 2024 (Online) 27th January Evening Shift

A closed organ pipe $150\ \text{cm}$ long gives 7 beats per second with an open organ pipe of length $350\ \text{cm}$, both vibrating in fundamental mode. The velocity of sound is _____ m/s.

Open solution

Q12. JEE Main 2023 (Online) 15th April Morning Shift

The fundamental frequency of vibration of a string stretched between two rigid supports is $50\ \text{Hz}$. The mass of the string is $18\ \text{g}$ and its linear mass density is $20\ \text{g/m}$. The speed of the transverse waves so produced in the string is _____ ms^{-1}

Open solution

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Q13. JEE Main 2023 (Online) 13th April Evening Shift

In an experiment with sonometer when a mass of $180\ \mathrm{g}$ is attached to the string, it vibrates with fundamental frequency of $30\ \mathrm{Hz}$. When a mass m is attached, the string vibrates with fundamental frequency of $50\ \mathrm{Hz}$. The value of m is _____ g.

Open solution

Q14. JEE Main 2023 (Online) 12th April Morning Shift

For a certain organ pipe, the first three resonance frequencies are in the ratio of $1 : 3 : 5$ respectively. If the frequency of fifth harmonic is $405\ \mathrm{Hz}$ and the speed of sound in air is $324\ \mathrm{ms}^{-1}$ the length of the organ pipe is _____ m.

Open solution

Q15. JEE Main 2023 (Online) 11th April Evening Shift

A wire of density $8 \times 10^3\ \mathrm{kg/m}^3$ is stretched between two clamps $0.5\ \mathrm{m}$ apart. The extension developed in the wire is $3.2 \times 10^{-4}\ \mathrm{m}$. If $Y = 8 \times 10^{10}\ \mathrm{N/m}^2$, the fundamental frequency of vibration in the wire will be _____ Hz.

Open solution

Q16. JEE Main 2023 (Online) 11th April Morning Shift

The equation of wave is given by

$\mathrm{Y} = 10^{-2} \sin 2\pi(160t - 0.5x + \pi/4)$

where x and Y are in m and t in s. The speed of the wave is _____ $\mathrm{km\ h}^{-1}$.

Open solution

Q17. JEE Main 2023 (Online) 10th April Morning Shift

A transverse harmonic wave on a string is given by

$y(x, t) = 5 \sin(6t + 0.003x)$

where x and y are in cm and t in sec. The wave velocity is _____ ms^{-1}.

Open solution

Q18. JEE Main 2023 (Online) 8th April Evening Shift

A guitar string of length 90 cm vibrates with a fundamental frequency of 120 Hz. The length of the string producing a fundamental frequency of 180 Hz will be _____ cm.

Open solution

Q19. JEE Main 2023 (Online) 8th April Morning Shift

An organ pipe $40\ \mathrm{cm}$ long is open at both ends. The speed of sound in air is $360\ \mathrm{ms}^{-1}$. The frequency of the second harmonic is _____ Hz.

Open solution

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Q20. JEE Main 2023 (Online) 31st January Evening Shift

The displacement equations of two interfering waves are given by

$y_1 = 10\sin\left(\omega t + \frac{\pi}{3}\right)\text{cm}, y_2 = 5[\sin\omega t + \sqrt{3}\cos\omega t]\text{cm}$ respectively.

The amplitude of the resultant wave is _____ cm.

Open solution

Q21. JEE Main 2023 (Online) 29th January Morning Shift

Two simple harmonic waves having equal amplitudes of 8 cm and equal frequency of 10 Hz are moving along the same direction. The resultant amplitude is also 8 cm. The phase difference between the individual waves is _____ degree.

Open solution

Q22. JEE Main 2023 (Online) 25th January Morning Shift

The distance between two consecutive points with phase difference of 60° in a wave of frequency 500 Hz is 6.0 m. The velocity with which wave is travelling is _____ km/s

Open solution

Q23. JEE Main 2022 (Online) 27th July Evening Shift

A wire of length 30 cm, stretched between rigid supports, has it's n^{th} and $(n + 1)^{th}$ harmonics at 400 Hz and 450 Hz, respectively. If tension in the string is 2700 N, it's linear mass density is _____ kg/m.

Open solution

Q24. JEE Main 2022 (Online) 29th June Evening Shift

In an experiment to determine the velocity of sound in air at room temperature using a resonance tube, the first resonance is observed when the air column has a length of 20.0 cm for a tuning fork of frequency 400 Hz is used. The velocity of the sound at room temperature is 336 ms^{-1}. The third resonance is observed when the air column has a length of _____ cm.

Open solution

Q25. JEE Main 2022 (Online) 28th June Evening Shift

A tuning fork of frequency 340 Hz resonates in the fundamental mode with an air column of length 125 cm in a cylindrical tube closed at one end. When water is slowly poured in it, the minimum height of water required for observing resonance once again is _____ cm.

(Velocity of sound in air is 340 ms^{-1})

Open solution

Q26. JEE Main 2022 (Online) 26th June Evening Shift

A set of 20 tuning forks is arranged in a series of increasing frequencies. If each fork gives 4 beats with respect to the preceding fork and the frequency of the last fork is twice the frequency of the first, then the frequency of last fork is _____ Hz.

Open solution

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Q27. JEE Main 2022 (Online) 25th June Morning Shift

The first overtone frequency of an open organ pipe is equal to the fundamental frequency of a closed organ pipe. If the length of the closed organ pipe is 20 cm, the length of the open organ pipe is _____ cm.

Open solution

Q28. JEE Main 2022 (Online) 24th June Evening Shift

Two travelling waves of equal amplitudes and equal frequencies move in opposite directions along a string. They interfere to produce a stationary wave whose equation is given by
$y = (10 \cos \pi x \sin \frac{2\pi t}{T})$ cm
The amplitude of the particle at $x = \frac{4}{3}$ cm will be _____ cm.

Open solution

Q29. JEE Main 2021 (Online) 31st August Morning Shift

A wire having a linear mass density 9.0×10^{-4} kg/m is stretched between two rigid supports with a tension of 900 N. The wire resonates at a frequency of 500 Hz. The next higher frequency at which the same wire resonates is 550 Hz. The length of the wire is _____ m.

Open solution

Q30. JEE Main 2021 (Online) 27th August Evening Shift

A tuning fork is vibrating at 250 Hz. The length of the shortest closed organ pipe that will resonate with the tuning fork will be _____ cm. (Take speed of sound in air as 340 ms^{-1})

Open solution

Q31. JEE Main 2021 (Online) 26th August Evening Shift

Two waves are simultaneously passing through a string and their equations are :

$y_1 = A_1 \sin k(x - vt)$, $y_2 = A_2 \sin k(x - vt + x_0)$. Given amplitudes A_1 = 12 mm and A_2 = 5 mm, x_0 = 3.5 cm and wave number k = 6.28 cm^{-1}. The amplitude of resulting wave will be ..... mm.

Open solution

Q32. JEE Main 2021 (Online) 26th August Morning Shift

Two travelling waves produce a standing wave represented by equation,

y = 1.0 mm cos(1.57 cm^{-1}) x sin(78.5 s^{-1})t.

The node closest to the origin in the region x > 0 will be at x = ... cm.

Open solution

Q33. JEE Main 2021 (Online) 20th July Morning Shift

The amplitude of wave disturbance propagating in the positive x-direction is given by $y = \frac{1}{1+x^2}$ at time t = 0 and $y = \frac{1}{1+(x-2)^2}$ at t = 1 s, where x and y are in metres. The shape of wave does not change during the propagation. The velocity of the wave will be _____ m/s.

Open solution

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Q34. JEE Main 2021 (Online) 16th March Evening Shift

A closed organ pipe of length L and an open organ pipe contain gases of densities ρ_1 and ρ_2 respectively. The compressibility of gases are equal in both the pipes. Both the pipes are vibrating in their first overtone with same frequency. The length of the open pipe is $\frac{x}{3}L\sqrt{\frac{\rho_1}{\rho_2}}$ where x is _____. (Round off to the Nearest Integer)

Open solution

Q35. JEE Main 2021 (Online) 26th February Morning Shift

The mass per unit length of a uniform wire is 0.135 g/cm. A transverse wave of the form y = – 0.21 sin (x + 30t) is produced in it, where x is in meter and t is in second. Then, the expected value of tension in the wire is x $\times$ 10^{-2} N. Value of x is _____. (Round off to the nearest integer)

Open solution

Q36. JEE Main 2021 (Online) 25th February Evening Shift

The percentage increase in the speed of transverse waves produced in a stretched string if the tension is increased by 4%, will be _____%.

Open solution

Q37. JEE Main 2020 (Online) 2nd September Evening Slot

A wire of density 9 $\times$ 10^{-3} kg cm^{-3} is stretched between two clamps 1 m apart. The resulting strain in the wire is 4.9 $\times$ 10^{-4}. The lowest frequency of the transverse vibrations in the wire is : (Young's modulus of wire Y = 9 $\times$ 10^{10} Nm^{-2}), (to the nearest integer), _____

Open solution

Q38. JEE Main 2020 (Online) 8th January Morning Slot

A one metre long (both ends open) organ pipe is kept in a gas that has double the density of air at STP. Assuming the speed of sound in air at STP is 300 m/s, the frequency difference between the fundamental and second harmonic of this pipe is _____ Hz.

Open solution

MCQ (Single Correct Answer) 97 questions

Q39. JEE Main 2026 (Online) 2nd April Morning Shift

The equation of a plane progressive wave is given by $y = 5\cos\pi\left(200t - \frac{x}{150}\right)$ where x and y are in cm and t is in second. The velocity of the wave is _____ m/s.

(A) 120 (B) 150 (C) 200 (D) 300

Open solution

Q40. JEE Main 2026 (Online) 24th January Evening Shift

The fifth harmonic of a closed organ pipe is found to be in unison with the first harmonic of an open pipe. The ratio of lengths of closed pipe to that of the open pipe is $5/x$. The value of x is ____

(A) 3 (B) 2 (C) 4 (D) 1

Open solution

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Q41. JEE Main 2026 (Online) 24th January Evening Shift

A point source is kept at the center of a spherically enclosed detector. If the volume of the detector increased by 8 times, the intensity will

(A) increase by 64 times
(B) decrease by 4 times
(C) decrease by 8 times
(D) increase by 8 times

Open solution

Q42. JEE Main 2026 (Online) 22nd January Evening Shift

In an open organ pipe ν_3 and ν_6 are 3rd and 6th harmonic frequencies, respectively. If $\nu_6 - \nu_3 = 2200\ \text{Hz}$ then length of the pipe is ____ mm .

(Take velocity of sound in air is $330\ \text{m/s}$.)

(A) 200
(B) 225
(C) 275
(D) 250

Open solution

Q43. JEE Main 2026 (Online) 21st January Morning Shift

Two strings (A, B) having linear densities $\mu_A = 2 \times 10^{-4}\ \text{kg/m}$ and, $\mu_B = 4 \times 10^{-4}\ \text{kg/m}$ and lengths $L_A = 2.5\ \text{m}$ and $L_B = 1.5\ \text{m}$ respectively are joined. Free ends of A and B are tied to two rigid supports C and D, respectively creating a tension of 500 N in the wire. Two identical pulses, sent from C and D ends, take time t_1 and t_2, respectively, to reach the joint. The ratio t_1/t_2 is:

(A) 1.90
(B) 1.18
(C) 1.08
(D) 1.67

Open solution

Q44. JEE Main 2025 (Online) 8th April Evening Shift

The amplitude and phase of a wave that is formed by the superposition of two harmonic travelling waves, $y_1(x,t) = 4\sin(kx - \omega t)$ and $y_2(x,t) = 2\sin(kx - \omega t + \frac{2\pi}{3})$, are:

(Take the angular frequency of initial waves same as ω)

(A) $\left[\sqrt{3}, \frac{\pi}{6}\right]$
(B) $\left[2\sqrt{3}, \frac{\pi}{6}\right]$
(C) $\left[6, \frac{2\pi}{3}\right]$
(D) $\left[6, \frac{\pi}{3}\right]$

Open solution

Q45. JEE Main 2025 (Online) 8th April Evening Shift

Two strings with circular cross section and made of same material, are stretched to have same amount of tension. A transverse wave is then made to pass through both the strings. The velocity of the wave in the first string having the radius of cross section R is v_1, and that in the other string having radius of cross section R/2 is v_2. Then $\frac{v_2}{v_1}$ =

(A) 8
(B) 4
(C) 2
(D) $\sqrt{2}$

Open solution

Q46. JEE Main 2025 (Online) 7th April Evening Shift

The equation of a wave travelling on a string is y = sin[20πx + 10πt], where x and t are distance and time in SI units. The minimum distance between two points having the same oscillating speed is :

(A) 10 cm
(B) 2.5 cm
(C) 20 cm
(D) 5.0 cm

Open solution

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Q47. JEE Main 2025 (Online) 7th April Morning Shift

Two harmonic waves moving in the same direction superimpose to form a wave $x = \mathrm{a}\cos(1.5\mathrm{t})\cos(50.5\mathrm{t})$ where t is in seconds. Find the period with which they beat. (close to nearest integer)

Open solution

(A) 1 s (B) 4 s (C) 2 s (D) 6 s

Q48. JEE Main 2025 (Online) 4th April Evening Shift

Displacement of a wave is expressed as $x(t) = 5\cos\left(628t + \frac{\pi}{2}\right)\mathrm{m}$. The wavelength of the wave when its velocity is $300\ \mathrm{m/s}$ is :

$(\pi = 3.14)$

Open solution

(A) 0.33 m (B) 0.5 m (C) 3 m (D) 5 m

Q49. JEE Main 2025 (Online) 4th April Morning Shift

In an experiment with a closed organ pipe, it is filled with water by $\left(\frac{1}{5}\right)$ th of its volume. The frequency of the fundamental note will change by

Open solution

(A) 20% (B) 25% (C) -20% (D) -25%

Q50. JEE Main 2025 (Online) 3rd April Evening Shift

In the resonance experiment, two air columns (closed at one end) of 100 cm and 120 cm long, give 15 beats per second when each one is sounding in the respective fundamental modes. The velocity of sound in the air column is:

Open solution

(A) $370\ \mathrm{m/s}$ (B) $340\ \mathrm{m/s}$ (C) $335\ \mathrm{m/s}$ (D) $360\ \mathrm{m/s}$

Q51. JEE Main 2025 (Online) 2nd April Evening Shift

A sinusoidal wave of wavelength 7.5 cm travels a distance of 1.2 cm along the x-direction in 0.3 sec . The crest P is at $x = 0$ at $\mathrm{t} = 0\mathrm{sec}$ and maximum displacement of the wave is 2 cm . Which equation correctly represents this wave?

Open solution

(A) $y = 2\cos(0.83x - 3.35t)\mathrm{cm}$ (B) $y = 2\sin(0.83x - 3.5\mathrm{t})\mathrm{cm}$

(C) $y = 2\cos(0.13x - 0.5t)\mathrm{cm}$ (D) $y = 2\cos(3.35x - 0.83\mathrm{t})\mathrm{cm}$

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Q52. JEE Main 2025 (Online) 28th January Morning Shift

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason $\mathbf{R}$

Assertion A: A sound wave has higher speed in solids than gases.

Reason R: Gases have higher value of Bulk modulus than solids.

In the light of the above statements, choose the correct answer from the options given below

(A) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(B) $\mathbf{A}$ is false but $\mathbf{R}$ is true

(C) $\mathbf{A}$ is true but $\mathbf{R}$ is false

(D) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

Q53. JEE Main 2025 (Online) 23rd January Evening Shift

Open solution

The equation of a transverse wave travelling along a string is $y(x,t) = 4.0\sin\left[20 \times 10^{-3}x + 600t\right]\text{mm}$, where x is in mm and t is in second. The velocity of the wave is :

(A) $-60\ \text{m/s}$ (B) $+60\ \text{m/s}$ (C) $+30\ \text{m/s}$ (D) $-30\ \text{m/s}$

Q54. JEE Main 2025 (Online) 22nd January Morning Shift

Open solution

A closed organ and an open organ tube are filled by two different gases having same bulk modulus but different densities ρ_1 and ρ_2, respectively. The frequency of 9th harmonic of closed tube is identical with 4th harmonic of open tube. If the length of the closed tube is 10 cm and the density ratio of the gases is $\rho_1 : \rho_2 = 1 : 16$, then the length of the open tube is :

(A) $\frac{15}{7}$ cm (B) $\frac{20}{9}$ cm (C) $\frac{20}{7}$ cm (D) $\frac{15}{9}$ cm

Q55. JEE Main 2024 (Online) 8th April Evening Shift

Open solution

A plane progressive wave is given by $y = 2\cos 2\pi(330\text{t} - x)\text{m}$. The frequency of the wave is :

(A) 660 Hz (B) 340 Hz (C) 330 Hz (D) 165 Hz

Q56. JEE Main 2024 (Online) 31st January Evening Shift

Open solution

The speed of sound in oxygen at S.T.P. will be approximately: (given, $R = 8.3\ \text{JK}^{-1}, \gamma = 1.4$)

(A) 341 m/s (B) 333 m/s (C) 325 m/s (D) 315 m/s

Q57. JEE Main 2024 (Online) 31st January Morning Shift

Open solution

The fundamental frequency of a closed organ pipe is equal to the first overtone frequency of an open organ pipe. If length of the open pipe is $60\ \text{cm}$, the length of the closed pipe will be:

(A) 15 cm (B) 60 cm (C) 45 cm (D) 30 cm

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Q58. JEE Main 2023 (Online) 6th April Evening Shift

The ratio of speed of sound in hydrogen gas to the speed of sound in oxygen gas at the same temperature is:

(A) $1:1$ (B) $1:2$ (C) $1:4$ (D) $4:1$

Open solution

Q59. JEE Main 2023 (Online) 1st February Morning Shift

A steel wire with mass per unit length $7.0 \times 10^{-3}\,\mathrm{kg\,m^{-1}}$ is under tension of $70\,\mathrm{N}$. The speed of transverse waves in the wire will be:

(A) $10\,\mathrm{m/s}$ (B) $50\,\mathrm{m/s}$ (C) $100\,\mathrm{m/s}$ (D) $200\pi\,\mathrm{m/s}$

Open solution

Q60. JEE Main 2023 (Online) 24th January Morning Shift

A travelling wave is described by the equation

$y(x,t) = [0.05\sin(8x - 4t)]$ m

The velocity of the wave is : [all the quantities are in SI unit]

(A) $4\,\mathrm{ms^{-1}}$ (B) $2\,\mathrm{ms^{-1}}$ (C) $8\,\mathrm{ms^{-1}}$ (D) $0.5\,\mathrm{ms^{-1}}$

Open solution

Q61. JEE Main 2022 (Online) 28th July Morning Shift

In the wave equation

$y = 0.5\sin\frac{2\pi}{\lambda}(400\mathrm{t} - x)\,\mathrm{m}$

the velocity of the wave will be:

(A) 200 m/s (B) $200\sqrt{2}$ m/s (C) 400 m/s (D) $400\sqrt{2}$ m/s

Open solution

Q62. JEE Main 2022 (Online) 26th July Evening Shift

A transverse wave is represented by $y = 2\sin(\omega t - kx)\,\mathrm{cm}$. The value of wavelength (in cm) for which the wave velocity becomes equal to the maximum particle velocity, will be :

(A) 4π (B) 2π (C) π (D) 2

Open solution

Q63. JEE Main 2022 (Online) 30th June Morning Shift

Which of the following equations correctly represents a travelling wave having wavelength λ = 4.0 cm, frequency ν = 100 Hz and travelling in positive x-axis direction?

(A) $y = A\sin[(0.50\,\pi\,cm^{-1})x - (100\,\pi\,s^{-1})t]$

(B) $y = A\sin\ 2\pi[(0.25\ cm^{-1})x - (50\,s^{-1})t]$

(C) $y = A\sin\left[\left(\frac{2\pi}{4}\,cm^{-1}\right)x - \left(\frac{2\pi}{100}\,s^{-1}\right)t\right]$

(D) $y = A\sin\,\pi[(0.5\ cm^{-1})x - (200\ s^{-1})t]$

Open solution

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Q64. JEE Main 2022 (Online) 29th June Morning Shift

A longitudinal wave is represented by $x = 10 \sin 2\pi \left(nt - \frac{x}{\lambda}\right)$ cm. The maximum particle velocity will be four times the wave velocity if the determined value of wavelength is equal to :

(A) 2π (B) 5π (C) π (D) $\frac{5\pi}{2}$

Open solution

Q65. JEE Main 2022 (Online) 28th June Morning Shift

The velocity of sound in a gas, in which two wavelengths 4.08 m and 4.16 m produce 40 beats in 12s, will be :

(A) 282.8 ms^{-1} (B) 175.5 ms^{-1} (C) 353.6 ms^{-1} (D) 707.2 ms^{-1}

Open solution

Q66. JEE Main 2022 (Online) 27th June Evening Shift

If a wave gets refracted into a denser medium, then which of the following is true?

(A) wavelength, speed and frequency decreases.

(B) wavelength increases, speed decreases and frequency remains constant.

(C) wavelength and speed decreases but frequency remains constant.

(D) wavelength, speed and frequency increases.

Open solution

Q67. JEE Main 2022 (Online) 24th June Morning Shift

The equations of two waves are given by :

$y_1 = 5 \sin 2\pi(x - vt)$ cm

$y_2 = 3 \sin 2\pi(x - vt + 1.5)$ cm

These waves are simultaneously passing through a string. The amplitude of the resulting wave is :

(A) 2 cm (B) 4 cm (C) 5.8 cm (D) 8 cm

Open solution

Q68. JEE Main 2021 (Online) 17th March Evening Shift

A sound wave of frequency 245 Hz travels with the speed of 300 ms^{-1} along the positive x-axis. Each point of the wave moves to and fro through a total distance of 6 cm. What will be the mathematical expression of this travelling wave?

(A) $Y(x, t) = 0.03 [\sin 5.1x - (0.2 \times 10^3)t]$

(B) $Y(x, t) = 0.03 [\sin 5.1x - (1.5 \times 10^3)t]$

(C) $Y(x, t) = 0.06 [\sin 5.1x - (1.5 \times 10^3)t]$

(D) $Y(x, t) = 0.06 [\sin 0.8x - (0.5 \times 10^3)t]$

Open solution

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Q69. JEE Main 2021 (Online) 26th February Evening Shift

A tuning fork A of unknown frequency produces 5 beats/s with a fork of known frequency 340 Hz. When fork A is filed, the beat frequency decreases to 2 beats/s. What is the frequency of fork A?

(A) 335 Hz (B) 345 Hz (C) 338 Hz (D) 342 Hz

Open solution

Q70. JEE Main 2021 (Online) 25th February Morning Shift

A student is performing the experiment of resonance column. The diameter of the column tube is 6 cm. The frequency of the tuning fork is 504 Hz. Speed of the sound at the given temperature is 336 m/s. The zero of the metre scale coincides with the top end of the resonance column tube. The reading of the water level in the column when the first resonance occurs is :

(A) 13 cm (B) 18.4 cm (C) 16.6 cm (D) 14.8 cm

Open solution

Q71. JEE Main 2021 (Online) 24th February Evening Shift

Which of the following equations represents a travelling wave?

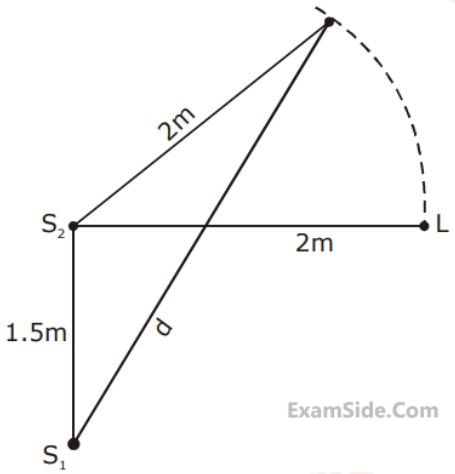
(A) $y = Ae^{x}\cos(\omega t - \theta)$ (B) $y = Ae^{-x^2}(vt + \theta)$

(C) $y = A\sin(15x - 2t)$ (D) $y = A\sin x\cos\omega t$

Open solution

Q72. JEE Main 2020 (Online) 5th September Evening Slot

Two coherent sources of sound, S_1 and S_2, produce sound waves of the same wavelength, $\lambda = 1$ m, in phase. S_1 and S_2 are placed 1.5 m apart (see fig). A listener, located at L, directly in front of S_2 finds that the intensity is at a minimum
when he is 2 m away from S_2. The listener moves away from S_1, keeping his distance from S_2 fixed. The adjacent maximum of intensity is observed when the listener is at a distance d from S_1. Then, d is :



(A) 12 m (B) 2 m (C) 3 m (D) 5 m

Open solution

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Q73. JEE Main 2020 (Online) 5th September Morning Slot

Assume that the displacement(s) of air is proportional to the pressure difference (Δp) created by a sound wave. Displacement (s) further depends on the speed of sound (v), density of air (ρ) and the frequency (f). If $\Delta p \sim 10$ Pa, v ~ 300 m/s, $\rho \sim 1$ kg/m^3 and f ~ 1000 Hz, then s will be of the order of (take the multiplicative constant to be 1) :

Open solution

(A) 1 mm (B) $\frac{3}{100}$ mm (C) 10 mm (D) $\frac{1}{10}$ mm

Q74. JEE Main 2020 (Online) 5th September Morning Slot

In a resonance tube experiment when the tube is filled with water up to a height of 17.0 cm from bottom, it resonates with a given tuning fork. When the water level is raised the next resonance with the same tuning fork occurs at a height of 24.5 cm. If the velocity of sound in air is 330 m/s, the tuning fork frequency is :

Open solution

(A) 2200 Hz (B) 3300 Hz (C) 1100 Hz (D) 550 Hz

Q75. JEE Main 2020 (Online) 4th September Morning Slot

For a transverse wave travelling along a straight line, the distance between two peaks (crests) is 5 m, while the distance between one crest and one trough is 1.5 m. The possible wavelengths (in m) of the wave are :

Open solution

(A) 1, 3, 5, ... (B) $\frac{1}{1}, \frac{1}{3}, \frac{1}{5}, ...$

(C) 1, 2, 3, ... (D) $\frac{1}{2}, \frac{1}{4}, \frac{1}{6},$

Q76. JEE Main 2020 (Online) 3rd September Morning Slot

A uniform thin rope of length 12 m and mass 6 kg hangs vertically from a rigid support and a block of mass 2 kg is attached to its free end. A transverse short wavetrain of wavelength 6 cm is produced at the lower end of the rope. What is the wavelength of the wavetrain (in cm) when it reaches the top of the rope ?

Open solution

(A) 12 (B) 3 (C) 9 (D) 6

Q77. JEE Main 2020 (Online) 2nd September Morning Slot

Two identical strings X and Z made of same material have tension T_X and T_Z in them. If their fundamental frequencies are 450 Hz and 300 Hz, respectively, then the ratio T_X/T_Z is

Open solution

(A) 2.25 (B) 0.44 (C) 1.25 (D) 1.5

Q78. JEE Main 2020 (Online) 9th January Evening Slot

A wire of length L and mass per unit length 6.0×10^{-3} kgm^{-1} is put under tension of 540 N. Two consecutive frequencies that it resonates at are : 420 Hz and 490 Hz. Then L in meters is :

Open solution

(A) 5.1 m (B) 2.1 m (C) 1.1 m (D) 8.1 m

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Q79. JEE Main 2020 (Online) 9th January Morning Slot

Three harmonic waves having equal frequency ν and same intensity I_0, have phase angles 0, $\frac{\pi}{4}$ and $-\frac{\pi}{4}$ respectively. When they are superimposed the intensity of the resultant wave is close to :

(A) 5.8 I_0 (B) 3 I_0 (C) 0.2 I_0 (D) I_0

Open solution

Q80. JEE Main 2020 (Online) 8th January Evening Slot

A transverse wave travels on a taut steel wire with a velocity of v when tension in it is 2.06×10^4 N. When the tension is changed to T, the velocity changed to v/2. The value of T is close to :

(A) 30.5×10^4 N (B) 2.50×10^4 N (C) 10.2×10^2 N (D) 5.15×10^3 N

Open solution

Q81. JEE Main 2020 (Online) 7th January Morning Slot

Speed of a transverse wave on a straight wire (mass 6.0 g, length 60 cm and area of cross-section 1.0 mm^2) is 90 ms^{-1}. If the Young's modulus of wire is 16×10^{11} Nm^{-2}, the extension of wire over its natural length is :

(A) 0.03 mm (B) 0.04 mm (C) 0.02 mm (D) 0.01 mm

Open solution

Q82. JEE Main 2019 (Online) 12th April Evening Slot

A tuning fork of frequency 480 Hz is used in an experiment for measuring speed of sound (v) in air by resonance tube method. Resonance is observed to occur at two successive lengths of the air column, l_1 = 30 cm and l_2 = 70 cm. Then, v is equal to -

(A) 338 ms^{-1} (B) 384 ms^{-1} (C) 379 ms^{-1} (D) 332 ms^{-1}

Open solution

Q83. JEE Main 2019 (Online) 12th April Evening Slot

A small speaker delivers 2 W of audio output. At what distance from the speaker will one detect 120 dB intensity sound ? [Given reference intensity of sound as 10^{-12} W/m^2]

(A) 20 cm (B) 10 cm (C) 40 cm (D) 30 cm

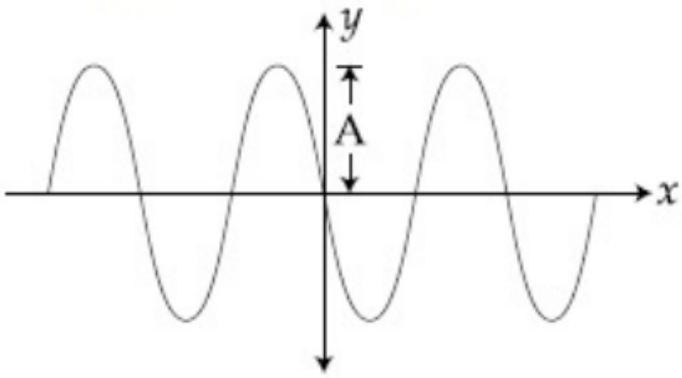
Open solution

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Q84. JEE Main 2019 (Online) 12th April Morning Slot

Open solution

A progressive wave travelling along the positive x-direction is represented by y(x,t) = Asin(kx − ωt + ϕ). Its snapshot at t = 0 is given in the figure.



For this wave, the phase ϕ is :

(A) $\frac{\pi}{2}$ (B) π (C) 0 (D) $-\frac{\pi}{2}$

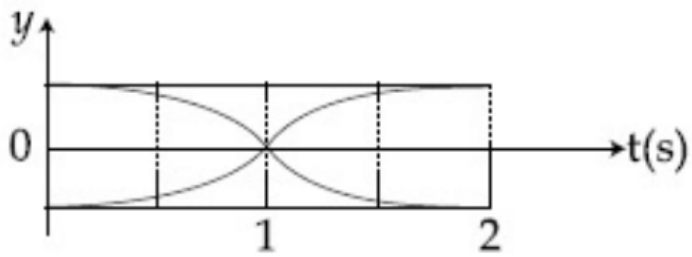
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Q85. JEE Main 2019 (Online) 10th April Evening Slot

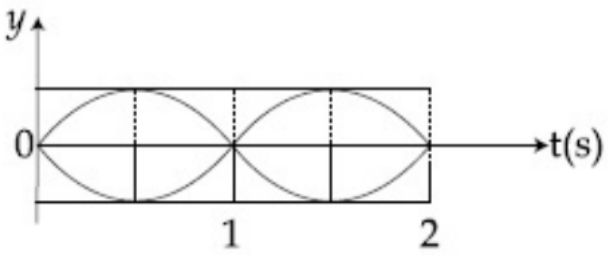
Open solution

The correct figure that shows, schematically, the wave pattern produced by superposition of two waves of frequencies 9 Hz and 11 Hz, is :

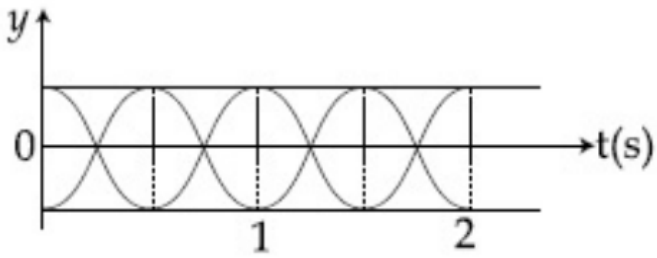
(A)



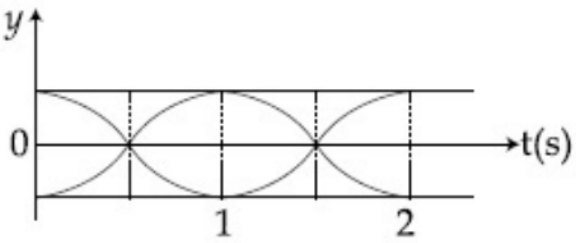
(B)



(C)



(D)



Q86. JEE Main 2019 (Online) 9th April Evening Slot

Open solution

A string 2.0 m long and fixed at its ends is driven by a 240 Hz vibrator. The string vibrates in its third harmonic mode. The speed of the wave and its fundamental frequency is :-

(A) 180m/s, 80 Hz (B) 180m/s, 120 Hz (C) 320m/s, 120 Hz (D) 320m/s, 80 Hz

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Q87. JEE Main 2019 (Online) 9th April Morning Slot

Open solution

The pressure wave, $P = 0.01 \sin [1000t - 3x]$ Nm^{-2}, corresponds to the sound produced by a vibrating blade on a day when atmospheric temperature is 0°C. On some other day, when temperature is T, the speed of sound produced by the same blade and at the same frequency is found to be 336 ms^{-1} . Approximate value of T is

(A) 12°C (B) 15°C (C) 4°C (D) 11°C

Q88. JEE Main 2019 (Online) 9th April Morning Slot

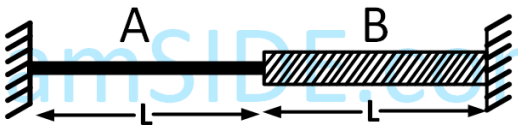
Open solution

A string is clamped at both the ends and it is vibrating in its 4th harmonic. The equation of the stationary wave is $Y = 0.3 \sin(0.157x) \cos(200\pi t)$. The length of the string is : (All quantities are in SI units.)

(A) 60 m (B) 20 m (C) 80 m (D) 40 m

Q89. JEE Main 2019 (Online) 8th April Morning Slot

Open solution



A wire of length 2L, is made by joining two wires A and B of same length but different radii r and 2r and made of the same material. It is vibrating at a frequency such that the joint of the two wires forms a node. If the number of antinodes in wire A is p and that in B is q then the ratio p : q is :

(A) 3 : 5 (B) 4 : 9 (C) 1 : 2 (D) 1 : 4

Q90. JEE Main 2019 (Online) 12th January Evening Slot

Open solution

A resonance tube is old and has jagged end. It is still used in the laboratory to determine velocity of sound in air. A tuning fork of frequency 512 Hz produces first resonance when the tube is filled with water to a mark 11 cm below a reference mark, near the open end of the tube. The experiment is repeated with another fork of frequency 256 Hz which produces first resonance when water reaches a mark 27 cm below the reference mark. The velocity of sound in air, obtained in the experiment, is close to :

(A) 335 ms^{-1} (B) 328 ms^{-1} (C) 341 ms^{-1} (D) 322 ms^{-1}

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Q91. JEE Main 2019 (Online) 12th January Morning Slot

A travelling harmonic wave is represented by the equation $y(x,t) = 10^{-3}\sin(50t + 2x)$, where, x and y are in metre and t is in seconds. Which of the following is a correct statement about the wave ?

(A) The wave is propagating along the positive x-axis with speed 100 ms^{-1}

(B) The wave is propagating along the positive x-axis with speed 25 ms^{-1}

(C) The wave is propagating along the negative x-axis with speed 25 ms^{-1}

(D) The wave is propagating along the negative x-axis with speed 100 ms^{-1}

Open solution

Q92. JEE Main 2019 (Online) 11th January Morning Slot

Equation of travelling wave on a stretched string of linear density 5 g/m is y = 0.03 sin(450 t – 9x) where distance and time are measured in SI units. The tension in the string is :

(A) 10 N (B) 7.5 N (C) 5 N (D) 12.5 N

Open solution

Q93. JEE Main 2019 (Online) 10th January Morning Slot

A string of length 1 m and mass 5 g is fixed at both ends. The tension in the string is 8.0 N. The string is set into vibration using an external vibrator of frequency 100 Hz. The separation between successive nodes on the string is close to -

(A) 16.6 cm (B) 10.0 cm (C) 20.0 cm (D) 33.3 cm

Open solution

Q94. JEE Main 2019 (Online) 9th January Morning Slot

A heavy ball of mass M is suspendeed from the ceiling of a car by a light string of mass m (m < < M). When the car is at rest, the speed of transverse waves in the string is 60 ms^{-1}. When the car has acceleration a, the wave-speed increases to 60.5 ms^{-1}. The value of a, in terms of gravitational acceleration g, is closest to :

(A) $\frac{g}{30}$ (B) $\frac{g}{5}$ (C) $\frac{g}{10}$ (D) $\frac{g}{20}$

Open solution

Q95. JEE Main 2018 (Online) 16th April Morning Slot

The end correction of a resonance column is 1 cm. If the shortest length resonating with the tuning fork is 10 cm, the next resonating length should be :

(A) 28 cm (B) 32 cm (C) 36 cm (D) 40 cm

Open solution

Q96. JEE Main 2018 (Offline)

A granite rod of 60 cm length is clamped at its middle point and is set into longitudinal vibrations. The density of granite is 2.7×10^3 kg/m^3 and its Young's modulus is 9.27×10^{10} Pa. What will be the fundamental frequency of the longitudinal vibrations ?

(A) 7.5 kHz (B) 5 kHz (C) 2.5 kHz (D) 10 kHz

Open solution

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Q97. JEE Main 2018 (Online) 15th April Evening Slot

5 beats / second are heard when a tuning fork is sounded with a sonometer wire under tension, when the length of the sonometer wire is either 0.95 m or 1 m. The frequency of the fork will be :

(A) 195 Hz (B) 150 Hz (C) 300 Hz (D) 251 Hz

Open solution

Q98. JEE Main 2018 (Online) 15th April Morning Slot

A tuning fork vibrates with frequency $256\ Hz$ and gives one beat per second with the third normal mode of vibration of an open pipe. What is the length of the pipe ? (Speed of sound in air is $340\ ms^{-1}$)

(A) $220\ cm$ (B) $190\ cm$ (C) $180\ cm$ (D) $200\ cm$

Open solution

Q99. JEE Main 2017 (Online) 9th April Morning Slot

A standing wave is formed by the superposition of two waves travelling in opposite directions. The transverse displacement is given by

$y(x, t) = 0.5 \sin\left(\frac{5\pi}{4}x\right) \cos(200\ \pi t)$.

What is the speed of the travelling wave moving in the positive x direction ?

(x and t are in meter and second, respectively.)

(A) 160 m/s (B) 90 m/s (C) 180 m/s (D) 120 m/s

Open solution

Q100. JEE Main 2017 (Online) 9th April Morning Slot

In an experiment to determine the period of a simple pendulum of length 1 m, it is attached to different spherical bobs of radii r_1 and r_2 . The two spherical bobs have uniform mass distribution. If the relative difference in the periods, is found to be 5×10^{-4} s, the difference in radii, $|r_1 - r_2|$ is best given by :

(A) 1 cm (B) 0.1 cm (C) 0.5 cm (D) 0.01 cm

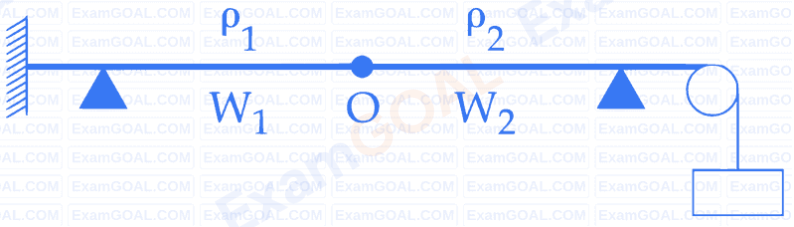
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Q101. JEE Main 2017 (Online) 8th April Morning Slot

Open solution

Two wires W_1 and W_2 have the same radius r and respective densities ρ_1 and ρ_2 such that $\rho_2 = 4\rho_1$. They are joined together at the point O, as shown in the figure. The combination is used as a sonometer wire and kept under tension T. The point O is midway between the two bridges. When a stationary wave is set up in the composite wire, the joint is found to be a node. The ratio of the number of antinodes formed in W_1 to W_2 is :



(A) 1 : 1 (B) 1 : 2 (C) 1 : 3 (D) 4 : 1

Q102. JEE Main 2016 (Offline)

Open solution

A uniform string of length $20\ m$ is suspended from a rigid support. A short wave pulse is introduced at its lowest end. It starts moving up the string. The time taken to reach the supports is :
(take $g = 10ms^{-2}$)

(A) $2\sqrt{2}s$ (B) $2\pi\sqrt{2}s$ (C) $2\pi\sqrt{2}s$ (D) $2\ s$

Q103. JEE Main 2016 (Offline)

Open solution

A pipe open at both ends has a fundamental frequency f in air. The pipe is dipped vertically in water so that half of it is in water. The fundamental frequency of the air column is now :

(A) $2f$ (B) f (C) $\frac{f}{2}$ (D) $\frac{3f}{4}$

Q104. JEE Main 2014 (Online) 19th April Morning Slot

Open solution

The total length of a sonometer wire between fixed ends is 110 cm. Two bridges are placed to divide the length of wire in ratio 6 : 3 : 2. The tension in the wire is 400 N and the mass per unit length is 0.01 kg/m. What is the minimum common frequency with which three parts can vibrate ?

(A) 1100 Hz (B) 1000 Hz (C) 166 Hz (D) 100 Hz

Q105. JEE Main 2014 (Online) 9th April Morning Slot

Open solution

A transverse wave is represented by :

$y = \frac{10}{\pi} \sin\left(\frac{2\pi}{T}t - \frac{2\pi}{\lambda}x\right)$

For what value of the wavelength the wave velocity is twice the maximum particle velocity ?

(A) 40 cm (B) 20 cm (C) 10 cm (D) 60 cm

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Q106. JEE Main 2014 (Offline)

A pipe of length $85\ cm$ is closed from one end. Find the number of possible natural oscillations of air column in the pipe whose frequencies lie below $1250\ Hz$. The velocity of sound in air is $340\ m/s$.

(A) 12 (B) 8 (C) 6 (D) 4

Open solution

Q107. JEE Main 2013 (Online) 25th April Morning Slot

In a transverse wave the distance between a crest and neighbouring trough at the same instant is 4.0 cm and the distance between a crest and trough at the same place is 1.0 cm. The next crest appears at the same place after a time interval of 0.4 s. The maximum speed of the vibrating particles in the medium is:

(A) $\frac{3\pi}{2}$ cm/s (B) $\frac{5\pi}{2}$ cm/s (C) $\frac{\pi}{2}$ cm/s (D) 2π cm/s

Open solution

Q108. JEE Main 2013 (Online) 23th April Morning Slot

A sonometer wire of length 114 cm is fixed at both the ends. Where should the two bridges be placed so as to divide the wire into three segments whose fundamental frequencies are in the ratio 1 : 3 : 4?

(A) At 36 cm and 84 cm from one end (B) At 24 cm and 72 cm from one end

(C) At 48 cm and 96 cm from one end (D) At 72 cm and 96 cm from one end

Open solution

Q109. JEE Main 2013 (Online) 9th April Morning Slot

When two sound waves travel in the same direction in a medium, the displacements of a particle located at 'x' at time 't' is given by :

$y_1 = 0.05\cos(0.50\pi x - 100\pi t)$

$y_2 = 0.05\cos(0.46\pi x - 92\pi t)$

where y_1, y_2 and x are in meters and t in seconds. The speed of sound in the medium is :

(A) 92 m/s (B) 200 m/s (C) 100 m/s (D) 332 m/s

Open solution

Q110. JEE Main 2013 (Online) 9th April Morning Slot

An engine approaches a hill with a constant speed. When it is at a distance of 0.9 km, it blows a whistle whose echo is heard by the driver after 5 seconds. If the speed of sound in air is 330 m/s, then the speed of the engine is :

(A) 32 m/s (B) 27.5 m/s (C) 60 m/s (D) 30 m/s

Open solution

Q111. JEE Main 2013 (Offline)

A sonometer wire of length $1.5\ m$ is made of steel. The tension in it produces an elastic strain of 1%. What is the fundamental frequency of steel if density and elasticity of steel are $7.7 \times 10^3\ kg/m^3$ and $2.2 \times 10^{11}\ N/m^2$ respectively ?

(A) $188.5\ Hz$ (B) $178.2\ Hz$ (C) $200.5\ Hz$ (D) $770\ Hz$

Open solution

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Q112. AIEEE 2012 (Online) 26th May

An air column in a pipe, which is closed at one end, will be in resonance with a vibrating tuning fork of frequency $264\ \text{Hz}$ if the length of the column in cm is (velocity of sound $= 330\ \text{m/s}$)

(A) 125.00 (B) 93.75 (C) 62.50 (D) 187.50

Open solution

Q113. AIEEE 2012 (Online) 26th May

The disturbance $y(x,t)$ of a wave propagating in the positive x-direction is given by $y = \dfrac{1}{1+x^2}$ at time $t = 0$ and by $y = \dfrac{1}{[1+(x-1)^2]}$ at $t = 2\ \text{s}$, where x and y are in meters. The shape of the wave disturbance does not change during the propagation. The velocity of wave in m/s is

(A) 2.0 (B) 4.0 (C) 0.5 (D) 1.0

Open solution

Q114. AIEEE 2012 (Online) 19th May

A uniform tube of length $60.5\ \text{cm}$ is held vertically with its lower end dipped in water. A sound source of frequency $500\ \text{Hz}$ sends sound waves into the tube. When the length of tube above water is $16\ \text{cm}$ and again when it is $50\ \text{cm}$, the tube resonates with the source of sound. Two lowest frequencies (in Hz), to which tube will resonate when it is taken out of water, are (approximately).

(A) 281, 562 (B) 281, 843 (C) 276, 552 (D) 272, 544

Open solution

Q115. AIEEE 2012 (Online) 12th May

A wave represented by the equation $y_1 = a\cos(kx - \omega t)$ is superimposed with another wave to form a stationary wave such that the point $x = 0$ is node. The equation for the other wave is

(A) $a\cos(kx - \omega t + \pi)$ (B) $a\cos(kx + \omega t + \pi)$

(C) $a\cos\left(kx + \omega t + \frac{\pi}{2}\right)$ (D) $a\cos\left(kx - \omega t + \frac{\pi}{2}\right)$

Open solution

Q116. AIEEE 2012 (Online) 7th May

Following are expressions for four plane simple harmonic waves

(i) $y_1 = A\cos 2\pi\left(n_1 t + \dfrac{x}{\lambda_1}\right)$

(ii) $y_2 = A\cos\left[2\pi\left(n_1 t + \dfrac{x}{\lambda_1}\right) + \pi\right]$

(iii) $y_3 = A\cos 2\pi\left(n_2 t + \dfrac{x}{\lambda_2}\right)$

(iv) $y_4 = A\cos 2\pi\left(n_2 t - \dfrac{x}{\lambda_2}\right)$

The pairs of waves which will produce destructive interference and stationary waves respectively in a medium, are

(A) (iii, iv), (i, ii) (B) (i, iii), (ii, iv)

(C) (i, iv), (ii, iii) (D) (i, ii), (iii, iv)

Open solution

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Q117. AIEEE 2012 (Online) 7th May

Open solution

This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
Statement 1: In the resonance tube experiment, if the tuning fork is replaced by another identical tuning fork but with its arm having been filed, the length of the air column should be increased to obtain resonance again.
Statement 2: On filing the arms, the frequency of a tuning fork increases.

(A) Statement 1 is true, Statement 2 is true, Statement 2 is a correct explanation of Statement 1.

(B) Statement 1 is true, Statement 2 is false.

(C) Statement 1 is false, Statement 2 is true.

(D) Statement 1 is true, Statement 2 is true, Statement 2 is not a correct explanation of Statement 1.

Q118. AIEEE 2012

Open solution

A cylindrical tube, open at both ends, has a fundamental frequency, f, in air. The tube is dipped vertically in water so that half of it is in water. The fundamental frequency of the air-column is now :

(A) f (B) $f/2$ (C) $3/4$ (D) $2f$

Q119. AIEEE 2011

Open solution

The transverse displacement $y(x,t)$ of a wave on a string is given by $y(x,t) = e^{-\left(ax^2+bt^2+2\sqrt{ab}\,xt\right)}$.
This represents a:

(A) wave moving in $-x$ direction with speed $\sqrt{\frac{b}{a}}$

(B) standing wave of frequency $\sqrt{b}$

(C) standing wave of frequency $\frac{1}{\sqrt{b}}$

(D) wave moving in $+x$ direction with speed $\sqrt{\frac{a}{b}}$

Q120. AIEEE 2010

Open solution

The equation of a wave on a string of linear mass density $0.04\ kg\,m^{-1}$ is given by $y = 0.02\,(m)\,\sin\left[2\pi\left(\frac{t}{0.04(s)} - \frac{x}{0.50(m)}\right)\right]$.
The tension in the string is

(A) $4.0N$ (B) $12.5\ N$ (C) $0.5\ N$ (D) $6.25\ N$

Q121. AIEEE 2009

Open solution

Three sound waves of equal amplitudes have frequencies $(v-1)$, v, $(v+1)$. They superpose to give beats. The number of beats produced per second will be :

(A) 3 (B) 2 (C) 1 (D) 4

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Q122. AIEEE 2008

While measuring the speed of sound by performing a resonance column experiment, a student gets the first resonance condition at a column length of $18\ cm$ during winter. Repeating the same experiment during summer, she measures the column length to be $x\ cm$ for the second resonance. Then

Open solution

(A) $18 > x$ (B) $x > 54$ (C) $54 > x > 36$ (D) $36 > x > 18$

Q123. AIEEE 2008

A wave travelling along the x-axis is described by the equation $y(x,t) = 0.005\ \cos\ (\alpha x - \beta t)$. If the wavelength and the time period of the wave are $0.08\ m$ and $2.0s$, respectively, then α and β in appropriate units are

Open solution

(A) $\alpha = 25.00\pi,\ \beta = \pi$ (B) $\alpha = \frac{0.08}{\pi},\ \beta = \frac{2.0}{\pi}$

(C) $\alpha = \frac{0.04}{\pi},\ \beta = \frac{1.0}{\pi}$ (D) $\alpha = 12.50\pi,\ \beta = \frac{\pi}{2.0}$

Q124. AIEEE 2007

A sound absorber attenuates the sound level by $20\ dB$. The intensity decreases by a factor of

Open solution

(A) 100 (B) 1000 (C) 10000 (D) 10

Q125. AIEEE 2006

A string is stretched between fixed points separated by $75.0\ cm$. It is observed to have resonant frequencies of $420\ Hz$ and $315\ Hz$. There are no other resonant frequencies between these two. Then, the lowest resonant frequency for this string is

Open solution

(A) $105\ Hz$ (B) $1.05\ Hz$ (C) $1050\ Hz$ (D) $10.5\ Hz$

Q126. AIEEE 2005

When two tuning forks (fork 1 and fork 2) are sounded simultaneously, 4 beats per second are heated. Now, some tape is attached on the prong of the fork 2. When the tuning forks are sounded again, 6 beats per second are heard. If the frequency of fork 1 is $200\ Hz$, then what was the original frequency of fork 2 ?

Open solution

(A) $202\ Hz$ (B) $200\ Hz$ (C) $204\ Hz$ (D) $196\ Hz$

Q127. AIEEE 2004

The displacement y of a particle in a medium can be expressed as, $y = 10^{-6}\ \sin\left(100t + 20x + \frac{\pi}{4}\right)$ m where t is in second and x in meter. The speed of the wave is

Open solution

(A) $20\ m/s$ (B) $5\,m/s$ (C) $2000\,m/s$ (D) $5\,\pi\,m/s$

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Q128. AIEEE 2003

The displacement y of a wave travelling in the x-direction is given by

$$y = 10^{-4} \sin\left(600t - 2x + \frac{\pi}{3}\right) \ metres$$

where x is expressed in metres and t in seconds. The speed of the wave - motion, in ms^{-1}, is

(A) 300 (B) 600 (C) 1200 (D) 200

Open solution

Q129. AIEEE 2003

A metal wire of linear mass density of $9.8\ g/m$ is stretched with a tension of $10\ kg\text{-}wt$ between two rigid supports 1 metre apart. The wire passes at its middle point between the poles of a permanent magnet, and it vibrates in resonance when carrying an alternating current of frequency n. The frequency n of the alternating source is

(A) $50\ Hz$ (B) $100\ Hz$ (C) $200\ Hz$ (D) $25\ Hz$

Open solution

Q130. AIEEE 2003

A tuning fork of known frequency $256\ Hz$ makes 5 beats per second with the vibrating string of a piano. The beat frequency decreases to 2 beats per second when the tension in the piano string is slightly increased. The frequency of the piano string before increasing the tension was

(A) $256 + 2Hz$ (B) $256 - 2Hz$ (C) $256 - 5Hz$ (D) $256 + 5Hz$

Open solution

Q131. AIEEE 2002

length of a string tied to two rigid supports is $40\ cm$. Maximum length (wavelength in cm) of a stationary wave produced on it is

(A) 20 (B) 80 (C) 40 (D) 120

Open solution

Q132. AIEEE 2002

Tube A has both ends open while tube B has one end closed, otherwise they are identical. The ratio of fundamental frequency of tube A and B is

(A) $1:2$ (B) $1:4$ (C) $2:1$ (D) $4:1$

Open solution

Q133. AIEEE 2002

A tuning fork arrangement (pair) produces 4 beats/sec with one fork of frequency $288\ cps$. A little wax is placed on the unknown fork and it then produces 2 beats/sec. The frequency of the unknown fork is

(A) $286\ cps$ (B) $292\ cps$ (C) $294\ cps$ (D) $288\ cps$

Open solution

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Q134. AIEEE 2002

A wave $y = a \sin(\omega t - kx)$ on a string meets with another wave producing a node at $x = 0$. Then the equation of the unknown wave is

(A) $y = a \sin(\omega t + kx)$

(B) $y = -a \sin(\omega t + kx)$

(C) $y = a \sin(\omega t - kx)$

(D) $y = -a \sin(\omega t - kx)$

Open solution

Q135. AIEEE 2002

When temperature increases, the frequency of a tuning fork

(A) increases

(B) decreases

(C) remains same

(D) increases or decreases depending on the material

Open solution

Out of syllabus 41 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q136. JEE Main 2023 (Online) 6th April Morning Shift · **Numerical**

A person driving car at a constant speed of $15\ \mathrm{m/s}$ is approaching a vertical wall. The person notices a change of $40\ \mathrm{Hz}$ in the frequency of his car's horn upon reflection from the wall. The frequency of horn is _____ Hz.

(Given: Speed of sound : $330\ \mathrm{m/s}$)

Open solution

Q137. JEE Main 2023 (Online) 25th January Evening Shift · **Numerical**

A train blowing a whistle of frequency 320 Hz approaches an observer standing on the platform at a speed of 66 m/s. The frequency observed by the observer will be (given speed of sound = 330 ms^{-1}) _____ Hz.

Open solution

Q138. JEE Main 2022 (Online) 28th July Morning Shift · **Numerical**

The frequency of echo will be _____ Hz if the train blowing a whistle of frequency 320 Hz is moving with a velocity of 36 km/h towards a hill from which an echo is heard by the train driver. Velocity of sound in air is 330 m/s.

Open solution

Q139. JEE Main 2022 (Online) 26th July Morning Shift · **Numerical**

When a car is approaching the observer, the frequency of horn is $100\ \mathrm{Hz}$. After passing the observer, it is $50\ \mathrm{Hz}$. If the observer moves with the car, the frequency will be $\frac{x}{3}\ \mathrm{Hz}$ where $x =$ _____.

Open solution

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Q140. JEE Main 2022 (Online) 25th July Evening Shift · **Numerical**

Two waves executing simple harmonic motions travelling in the same direction with same amplitude and frequency are superimposed. The resultant amplitude is equal to the $\sqrt{3}$ times of amplitude of individual motions. The phase difference between the two motions is _____ (degree).

Open solution

Q141. JEE Main 2022 (Online) 25th July Morning Shift · **Numerical**

An observer is riding on a bicycle and moving towards a hill at $18\ \mathrm{kmh}^{-1}$. He hears a sound from a source at some distance behind him directly as well as after its reflection from the hill. If the original frequency of the sound as emitted by source is $640\ \mathrm{Hz}$ and velocity of the sound in air is $320\ \mathrm{m/s}$, the beat frequency between the two sounds heard by observer will be _____ Hz.

Open solution

Q142. JEE Main 2022 (Online) 30th June Morning Shift · **Numerical**

An employee of a factory moving away from his workplace by a car listens to the siren of the factory. He drives the car at the speed of 72 kmh^{-1} in the direction of wind which is blowing at 72 kmh^{-1} speed. Frequency of siren is 720 Hz. The employee hears an apparent frequency of _____ Hz.

(Assume speed of sound to be 340 ms^{-1})

Open solution

Q143. JEE Main 2021 (Online) 27th August Morning Shift · **Numerical**

Two cars X and Y are approaching each other with velocities 36 km/h and 72 km/h respectively. The frequency of a whistle sound as emitted by a passenger in car X, heard by the passenger in car Y is 1320 Hz. If the velocity of sound in air is 340 m/s, the actual frequency of the whistle sound produced is ..... Hz.

Open solution

Q144. JEE Main 2021 (Online) 26th August Morning Shift · **Numerical**

A source and a detector move away from each other in absence of wind with a speed of 20 m/s with respect to the ground. If the detector detects a frequency of 1800 Hz of the sound coming from the source, then the original frequency of source considering speed of sound in air 340 m/s will be ..... Hz.

Open solution

Q145. JEE Main 2021 (Online) 20th July Morning Shift · **Numerical**

The frequency of a car horn encountered a change from 400 Hz to 500 Hz, when the car approaches a vertical wall. If the speed of sound is 330 m/s. Then the speed of car is _____ km/h.

Open solution

Q146. JEE Main 2021 (Online) 24th February Evening Shift · **Numerical**

Two cars are approaching each other at an equal speed of 7.2 km/hr. When they see each other, both blow horns having frequency of 676 Hz. The beat frequency heard by each driver will be _____ Hz. [Velocity of sound in air is 340 m/s.]

Open solution

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Q147. JEE Main 2023 (Online) 11th April Evening Shift · MCQ (Single Correct Answer)

A car P travelling at $20\ \mathrm{ms}^{-1}$ sounds its horn at a frequency of $400\ \mathrm{Hz}$. Another car Q is travelling behind the first car in the same direction with a velocity $40\ \mathrm{ms}^{-1}$. The frequency heard by the passenger of the car Q is approximately [Take, velocity of sound $= 360\ \mathrm{ms}^{-1}$]

(A) 485 Hz (B) 514 Hz (C) 421 Hz (D) 471 Hz

Open solution

Q148. JEE Main 2023 (Online) 8th April Morning Shift · MCQ (Single Correct Answer)

The engine of a train moving with speed $10\ \mathrm{ms}^{-1}$ towards a platform sounds a whistle at frequency $400\ \mathrm{Hz}$. The frequency heard by a passenger inside the train is: (neglect air speed. Speed of sound in air $= 330\ \mathrm{ms}^{-1}$)

(A) 200 Hz (B) 412 Hz (C) 400 Hz (D) 388 Hz

Open solution

Q149. JEE Main 2023 (Online) 29th January Morning Shift · MCQ (Single Correct Answer)

A person observes two moving trains, 'A' reaching the station and 'B' leaving the station with equal speed of $30\ \mathrm{m/s}$. If both trains emit sounds with frequency $300\ \mathrm{Hz}$, (Speed of sound: $330\ \mathrm{m/s}$) approximate difference of frequencies heard by the person will be:

(A) 10 Hz (B) 55 Hz (C) 80 Hz (D) 33 Hz

Open solution

Q150. JEE Main 2022 (Online) 27th June Morning Shift · MCQ (Single Correct Answer)

An observer moves towards a stationary source of sound with a velocity equal to one-fifth of the velocity of sound. The percentage change in the frequency will be :

(A) 20% (B) 10% (C) 5% (D) 0%

Open solution

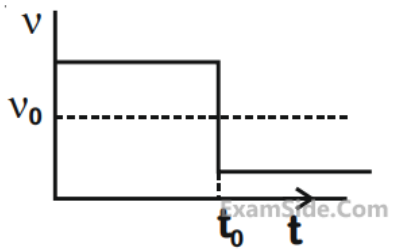
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Q151. JEE Main 2020 (Online) 6th September Morning Slot · MCQ (Single Correct Answer)

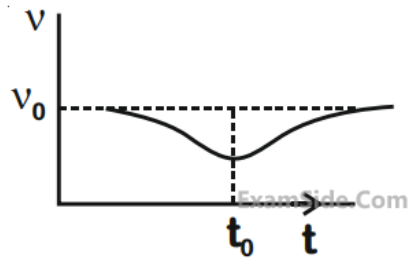
Open solution

A sound source S is moving along a straight track with speed v, and is emitting, sound of frequency v_0 (see figure). An observer is standing at a finite distance, at the point O, from the track. The time variation of frequency heard by the observer is best represented by:
(t_0 represents the instant when the distance between the source and observer is minimum)

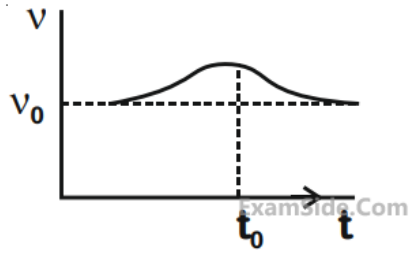
(A)



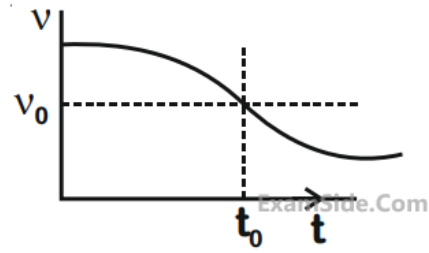
(B)



(C)



(D)



Q152. JEE Main 2020 (Online) 5th September Evening Slot · MCQ (Single Correct Answer)

Open solution

A driver in a car, approaching a vertical wall notices that the frequency of his car horn, has changed from 440 Hz to 480 Hz, when it gets reflected from the wall. If the speed of sound in air is 345 m/s, then the speed of the car is :

(A) 36 km/hr (B) 54 km/hr (C) 24 km/hr (D) 18 km/hr

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Q153. JEE Main 2020 (Online) 4th September Evening Slot · MCQ (Single Correct Answer)

The driver of a bus approaching a big wall notices that the frequency of his bus's horn changes from 420 Hz to 490 Hz when he hears it after it gets reflected from the wall. Find the speed of the bus if speed of the sound is 330 ms^{-1}.

Open solution

(A) 91 kmh^{-1} (B) 81 kmh^{-1} (C) 61 kmh^{-1} (D) 71 kmh^{-1}

Q154. JEE Main 2020 (Online) 7th January Evening Slot · MCQ (Single Correct Answer)

A stationary observer receives sound from two identical tuning forks, one of which approaches and the other one recedes with the same speed (much less than the speed of sound). The observer hears 2 beats/sec. The oscillation frequency of each tuning fork is v_0 = 1400 Hz and the velocity of sound in air is 350 m/s. The speed of each tuning fork is close to :

Open solution

(A) 1 m/s (B) $\frac{1}{8}$ m/s (C) $\frac{1}{4}$ m/s (D) $\frac{1}{2}$ m/s

Q155. JEE Main 2019 (Online) 12th April Evening Slot · MCQ (Single Correct Answer)

Two sources of sound S_1 and S_2 produce sound waves of same frequency 660 Hz. A listener is moving from source S_1 towards S_2 with a constant speed u m/s and he hears 10 beats/s. The velocity of sound is 330 m/s. Then, u equals :

Open solution

(A) 10.0 m/s (B) 5.5 m/s (C) 15.0 m/s (D) 2.5 m/s

Q156. JEE Main 2019 (Online) 12th April Morning Slot · MCQ (Single Correct Answer)

A submarine (A) travelling at 18 km/hr is being chased along the line of its velocity by another submarine (B) travelling at 27 km/hr. B sends a sonar signal of 500 Hz to detect A and receives a reflected sound of frequency υ. The value of υ is close to: (Speed of sound in water =1500 ms^{-1})

Open solution

(A) 507 Hz (B) 502 Hz (C) 499 Hz (D) 504 Hz

Q157. JEE Main 2019 (Online) 10th April Evening Slot · MCQ (Single Correct Answer)

A source of sound S is moving with a velocity of 50 m/s towards a stationary observer. The observer measures the frequency of the source as 1000 Hz. What will be the apparent frequency of the source when it is moving away from the observer after crossing him? (Take velocity of sound in air is 350 m/s)

Open solution

(A) 750 Hz (B) 857 Hz (C) 807 Hz (D) 1143 Hz

Q158. JEE Main 2019 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

A stationary source emits sound waves of frequency 500 Hz. Two observers moving along a line passing through the source detect sound to be of frequencies 480 Hz and 530 Hz. Their respective speeds are, in ms−1, (Given speed of sound = 300 m/s)

Open solution

(A) 12, 18 (B) 16, 14 (C) 12, 16 (D) 8, 18

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Q159. JEE Main 2019 (Online) 9th April Evening Slot · MCQ (Single Correct Answer)

Two cars A and B are moving away from each other in opposite directions. Both the cars are moving with a speed of 20 ms^{-1} with respect to the ground. If an observer in car A detects a frequency 2000 Hz of the sound coming from car B, what is the natural frequency of the sound source in car B ?

(speed of sound in air = 340 ms^{-1}) :-

(A) 2300 Hz (B) 2060 Hz (C) 2250 Hz (D) 2150 Hz

Open solution

Q160. JEE Main 2019 (Online) 10th January Evening Slot · MCQ (Single Correct Answer)

A closed organ pipe has a fundamental frequency of 1.5 kHz. The number of overtones that can be distinctly heard by a person with this organ pipe will be (Assume that the highest frequency a person can hear is 20,000 Hz)

(A) 4 (B) 7 (C) 6 (D) 5

Open solution

Q161. JEE Main 2019 (Online) 10th January Morning Slot · MCQ (Single Correct Answer)

A train moves towards a stationary observer with speed 34 m/s. The train sounds a whistle and its frequency registered by the observer is f_1. If the speed of the train is reduced to 17 m/s, the frequency registered is f_2. If speed of sound is 340 m/s, then the ratio f_1/f_2 is -

(A) 19/18 (B) 20/19 (C) 21/20 (D) 18/17

Open solution

Q162. JEE Main 2019 (Online) 9th January Evening Slot · MCQ (Single Correct Answer)

A musician using an open flute of length 50 cm producess second harmonic sound waves. A person runs towards the musician from another end of hall at a speed of 10 km/h. If the wave speed is 330 m/s, the frequency heard by the running person shall be close to :

(A) 666 Hz (B) 753 Hz (C) 500 Hz (D) 333 Hz

Open solution

Q163. JEE Main 2018 (Online) 16th April Morning Slot · MCQ (Single Correct Answer)

Two sitar strings, A and B, playing the note 'Dha' are slightly out of tune and produce beats of frequency 5 Hz. The tension of the string B s slightly increased and the beat frequency is found to decrease by 3Hz. If the frequency of A is 425 Hz, the original frequency of B is :

(A) 430 Hz (B) 420 Hz (C) 428 Hz (D) 422 Hz

Open solution

Q164. JEE Main 2017 (Offline) · MCQ (Single Correct Answer)

An observer is moving with half the speed of light towards a stationary microwave source emitting waves at frequency 10 GHz. What is the frequency of the microwave measured by the observer?

(speed of light = 3 $\times 10^8$ ms^{-1})

(A) 15.3 GHz (B) 10.1 GHz (C) 12.1 GHz (D) 17.3 GHz

Open solution

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Q165. JEE Main 2016 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

A toy-car, blowing its horn, is moving with a steady speed of 5 m/s, away from a wall. An observer, towards whom the toy car is moving, is able to hear 5 beats per second. If the velocity of sound in air is 340 m/s, the frequency of the horn of the toy car is close to :

Open solution

(A) 680 Hz (B) 510 Hz (C) 340 Hz (D) 170 Hz

Q166. JEE Main 2016 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

Two engines pass each other moving in opposite directions with uniform speed of 30 m/s. One of them is blowing a whistle of frequency 540 Hz. Calculate the frequency heard by driver of second engine before they pass each other. Speed of sound is 330 m/sec :

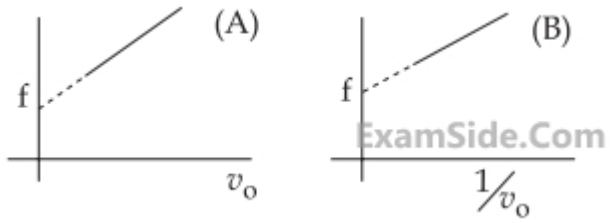
Open solution

(A) 450 Hz (B) 540 Hz (C) 648 Hz (D) 270 Hz

Q167. JEE Main 2015 (Online) 11th April Morning Slot · MCQ (Single Correct Answer)

A source of sound emits sound waves at frequency f_0. It is moving towards an observer with fixed speed v_s ($v_s < v$, where v is the speed of the sound in air). If the observer were to move towards the source with speed v_0, one of the following two graphs (A and B) will give the correct variation of the frequency f heard by the observer as v_0 is changed.

Open solution



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The variation of f with v_0 is given correctly by :

(A) graph A with slope = $\frac{f_0}{(v-v_s)}$

(B) graph A with slope = $\frac{f_0}{(v+v_s)}$

(C) graph B with slope = $\frac{f_0}{(v-v_s)}$

(D) graph B with slope = $\frac{f_0}{(v+v_s)}$

Q168. JEE Main 2015 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

A bat moving at 10 ms^{-1} towards a wall sends a sound signal of 8000 Hz towards it. On reflection it hears a sound of frequency f. The value of f in Hz is close to (speed of sound = 320 ms^{-1})

Open solution

(A) 8258 (B) 8516 (C) 8000 (D) 8424

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Q169. JEE Main 2015 (Offline) · MCQ (Single Correct Answer)

A train is moving on a straight track with speed $20\,ms^{-1}$. It is blowing its whistle at the frequency of $1000\,Hz$. The percentage change in the frequency heard by a person standing near the track as the train passes him is (speed of sound $= 320\,ms^{-1}$) close to :

Open solution

(A) 18% (B) 24% (C) 6% (D) 12%

Q170. JEE Main 2014 (Online) 12th April Morning Slot · MCQ (Single Correct Answer)

A source of sound A emitting waves of frequency 1800 Hz is falling towards ground with a terminal speed v. The observer B on the ground directly beneath the source receives waves of frequency 2150 Hz, The source A receives waves, reflected from ground, of frequency nearly : (Speed of sound = 343 m/s)

Open solution

(A) 2150 Hz (B) 2500 Hz (C) 1800 Hz (D) 2400 Hz

Q171. JEE Main 2014 (Online) 11th April Morning Slot · MCQ (Single Correct Answer)

Two factories are sounding their sirens at 800 Hz. A man goes from one factory to other at a speed of 2 m/s. The velocity of sound is 320 m/s. The number of beats heard by the person in one second will be :

Open solution

(A) 2 (B) 4 (C) 8 (D) 10

Q172. JEE Main 2013 (Online) 22th April Morning Slot · MCQ (Single Correct Answer)

A and B are two sources generating sound waves. A listener is situated at C. The frequency of the source at A is 500 Hz. A, now, moves towards C with a speed 4 m/s. The number of beats heard at C is 6. When A moves away from C with speed 4 m/s, the number of beats heard at C is 18. The speed of sound is 340 m/s. The frequency of the source at B is :

Open solution



(A) 500 Hz (B) 506 Hz (C) 512 Hz (D) 494 Hz

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Q173. AIEEE 2012 (Online) 12th May · MCQ (Single Correct Answer)

Open solution

This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
Statement 1: Bats emitting ultrasonic waves can detect the location of a prey by hearing the waves reflected from it.
Statement 2: When the source and the detector are moving, the frequency of reflected waves is changed.

(A) Statement 1 is false, Statement 2 is true.

(B) Statement 1 is true, Statement 2 is false.

(C) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1.

(D) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1.

Q174. AIEEE 2009 · MCQ (Single Correct Answer)

Open solution

A motor cycle starts from rest and accelerates along a straight path at $2m/s^2$. At the starting point of the motor cycle there is a stationary electric siren. How far has the motor cycle gone when the driver hears the frequency of the siren at 94% of its value when the motor cycle was at rest? (Speed of sound $= 330\,ms^{-1}$)

(A) $98\ m$ (B) $147\ m$ (C) $196\,m$ (D) $49\ m$

Q175. AIEEE 2006 · MCQ (Single Correct Answer)

Open solution

A whistle producing sound waves of frequencies $9500\ Hz$ and above is approaching a stationary person with speed $v\ ms^{-1}$. The velocity of sound in air is $300\,ms^{-1}$. If the person can hear frequencies upto a maximum of $10,000\ HZ$, the maximum value of v upto which he can hear whistle is

(A) $15\sqrt{2}\ ms^{-1}$ (B) $\frac{15}{\sqrt{2}}\ ms^{-1}$

(C) $15\ ms^{-1}$ (D) $30\ ms^{-1}$

Q176. AIEEE 2005 · MCQ (Single Correct Answer)

Open solution

An observer moves towards a stationary source of sound, with a velocity one-fifth of the velocity of sound. What is the percentage increase in the apparent frequency ?

(A) 0.5% (B) zero (C) 20% (D) 5%

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Answer Key

Numerical

1. **349**	2. **384**	3. **819**	4. **600**	5. **16**
6. **740**	7. **60**	8. **6**	9. **3**	10. **400**
11. **294**	12. **90**	13. **500**	14. **1**	15. **80**
16. **1152**	17. **20**	18. **60**	19. **900**	20. **20**
21. **120**	22. **18**	23. **3**	24. **104**	25. **50**
26. **152**	27. **80**	28. **5**	29. **10**	30. **34**
31. **7**	32. **1**	33. **2**	34. **4**	35. **1215**
36. **2**	37. **35**	38. **105.75 to 106.07**		

MCQ (Single Correct Answer)

39. **D**	40. **B**	41. **B**	42. **B**	43. **B**
44. **B**	45. **C**	46. **D**	47. **C**	48. **C**
49. **B**	50. **D**	51. **A**	52. **C**	53. **D**
54. **B**	55. **C**	56. **D**	57. **A**	58. **D**
59. **C**	60. **D**	61. **C**	62. **A**	63. **D**
64. **B**	65. **D**	66. **C**	67. **A**	68. **B**
69. **A**	70. **D**	71. **C**	72. **C**	73. **B**
74. **A**	75. **B**	76. **A**	77. **A**	78. **B**
79. **A**	80. **D**	81. **A**	82. **B**	83. **C**
84. **B**	85. **C**	86. **D**	87. **C**	88. **C**
89. **C**	90. **B**	91. **C**	92. **D**	93. **C**
94. **B**	95. **B**	96. **B**	97. **A**	98. **D**
99. **A**	100. **B**	101. **B**	102. **A**	103. **B**
104. **B**	105. **A**	106. **C**	107. **B**	108. **D**
109. **B**	110. **D**	111. **B**	112. **B**	113. **C**
114. **D**	115. **B**	116. **D**	117. **C**	118. **A**
119. **A**	120. **D**	121. **B**	122. **B**	123. **A**
124. **A**	125. **A**	126. **D**	127. **B**	128. **A**
129. **A**	130. **C**	131. **B**	132. **C**	133. **B**
134. **B**	135. **B**			

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Out of syllabus

136. **420**	137. **400**	138. **340**	139. **200**	140. **60**
141. **20**	142. **680**	143. **1210**	144. **2025**	145. **132**
146. **8**	147. **C**	148. **C**	149. **B**	150. **A**
151. **D**	152. **B**	153. **A**	154. **C**	155. **D**
156. **B**	157. **A**	158. **A**	159. **C**	160. **B**
161. **A**	162. **A**	163. **B**	164. **D**	165. **D**
166. **C**	167. **A**	168. **B**	169. **D**	170. **B**
171. **D**	172. **C**	173. **C**	174. **A**	175. **C**
176. **C**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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